2026

ETL PROJECT

Extract - Transform - Load

PROJECT NAME:
Supporting a
hotel chain

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Overview

Supporting a Parisian hotel chain in detecting nearby events in order to adjust its prices, its stock, its staffing...

Objective # 1

Collect data on events in Paris.

Objective # 2

Enrich and store the data in order to predict the popularity of events.

Objective # 3

Build an API so that the user can run the project on their own.



Extract - Data scraping

This website lists all current and upcoming music events. We scraped this site and retrieved all upcoming events in Paris.

Data transformation

Step #1

Data structuring

Step #2

Data enrichment to gather as much information as possible about the events

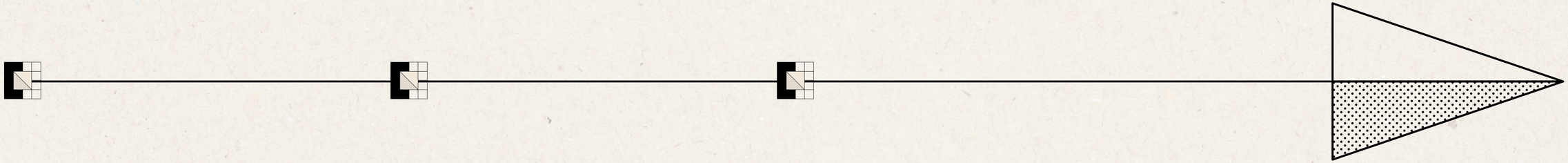
Step #3

Predicting the popularity of each upcoming day.



Load - Data ingestion into GCP

The enriched data is now stored in GCP
and our ETL pipeline is complete.



Step 1
Extract

Step 2
Transform

Step 3
Load

Building an API

TO MAKE THE PROJECT EASIER TO USE, AN API WAS BUILT IN PYTHON WITH FASTAPI. IT ALLOWS THE HOTEL CHAIN TO USE THE PROJECT DIRECTLY WITHOUT ANY ASSISTANCE. HERE IS WHAT IT CAN DO:

get an overview of all the upcoming events in the city.	get the number of concerts by popularity in the Paris region
get an overview of the number of events per day over a given week.	get the nationality of the artists performing in the Paris region
get the expected number of people in each city (and district) in the Paris region for each day.	get the names of the cities and districts in the database in order to search for the expected number of people
get events by artist and venue in order to anticipate attendance, for instance to adjust room prices in hotels near the venue...	run queries on the entire database using SQL query parameters

